

Steps	Reason
1. $\forall x (C(x) \Rightarrow DM(x))$	Premise
2. $DM(Natasha)$	Premise
3. $C(Natasha) \Rightarrow DM(Natasha)$	Universal instantiation on 1
4. $C(Natasha)$	Fallacy of affirming the conclusion on 2,3

This is an invalid argument.

c) All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.

Soln: $P(x)$: x is a parrot $F(x)$: x likes fruit.
 Domain: All birds.

Premise: 1) $\forall x (P(x) \Rightarrow F(x))$ 2. $\neg P(\text{Pet bird})$

Conclusion: $\neg F(\text{Pet bird})$

Steps	Reason
1. $\forall x (P(x) \Rightarrow F(x))$	Premise
2. $\neg P(\text{Pet bird})$	Premise
3. $P(\text{Pet bird}) \Rightarrow F(\text{Pet bird})$	Universal instantiation on 1
4. $\neg F(\text{Pet bird})$	Fallacy of denying the hypothesis on 3,2

This is an invalid argument.

d) Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.